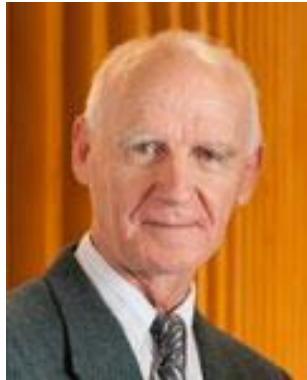


Use of Duval Triangles and Pentagons for the Interpretation of DGA in Transformers



Michel Duval, creator of the Duval Triangle and internationally renowned leader in the field of dissolved gas analysis (DGA) for transformer condition monitoring will offer new insights into the use of Duval Triangles and Pentagons for the interpretation of DGA.

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Interpretation of DGA Results

- DGA is among the most powerful and widely used methods for detecting failures and faults in transformers and for evaluating their condition in service.
- Over the years, several DGA diagnosis methods have been developed, using the different hydrocarbon gases formed (H_2 , CH_4 , C_2H_6 , C_2H_4 , C_2H_2).

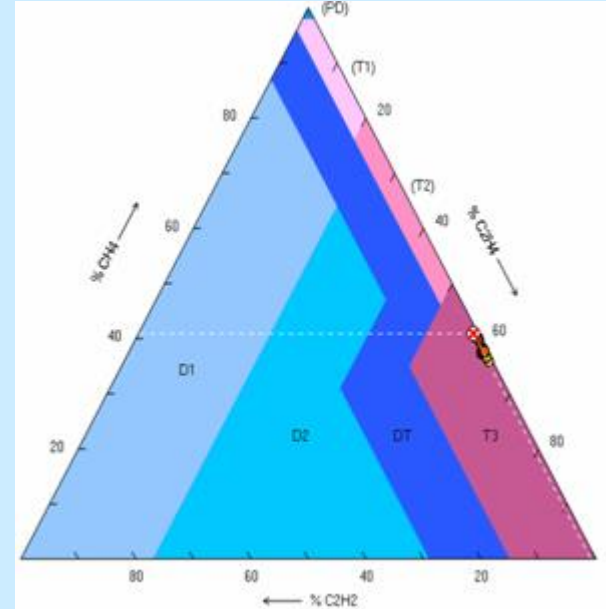
Diagnosis Methods Using Hydrocarbon Gases:

- One gas at a time (Key Gas Method).
- Two gases at a time (Rogers, IEC Ratio Methods).
- Three gases (Duval Triangles 1, 4, 5).
- Four gases (Korea).
- More recently, five gases (Duval Pentagons 1, 2).

Carbon oxides and furans are used to confirm diagnosis provided by hydrocarbon gases as to whether paper is involved or not in faults.

The Six Basic Faults Detectable by DGA:

- Using Rogers, IEC ratios and Duval Triangle 1
- PD: corona partial discharges
- D1: low-energy arcing
- D2: high-energy arcing
- T1: thermal faults $< 300^{\circ}\text{C}$
- T2: thermal faults $300 < T < 700^{\circ}\text{C}$
- T3: thermal faults $> 700^{\circ}\text{C}$

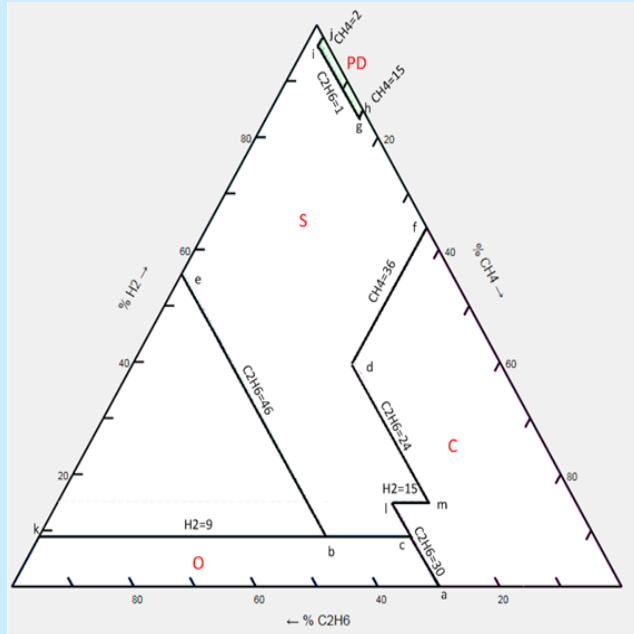


Triangle 1 (updated 2010), using
 CH_4 , C_2H_4 and C_2H_2

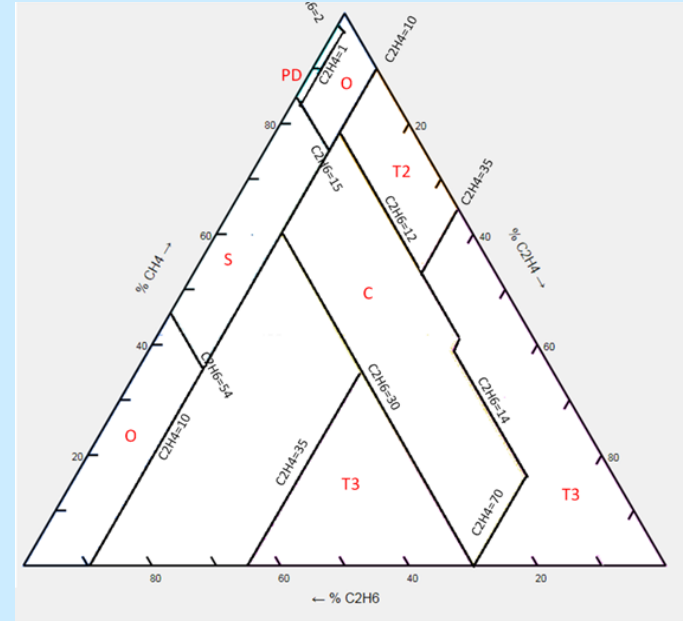
The Five Additional Sub-Types of Faults Detectable by DGA:

- Using Duval Triangles 4 and 5:
- S: stray gassing of oil $< 200^{\circ}\text{C}$
- O: overheating $< 250^{\circ}\text{C}$
- T2/T3 in oil only $> 300/ 700^{\circ}\text{C}$
- C: possibility of carbonization of paper $> 300^{\circ}\text{C}$
(with a probability of $\sim 80\%$, not 100%)
- PD: corona partial discharges (more precisely)

Duval Triangles 4 and 5 (updated 2012)



Triangle 4, using
 H_2 , CH_4 and C_2H_6

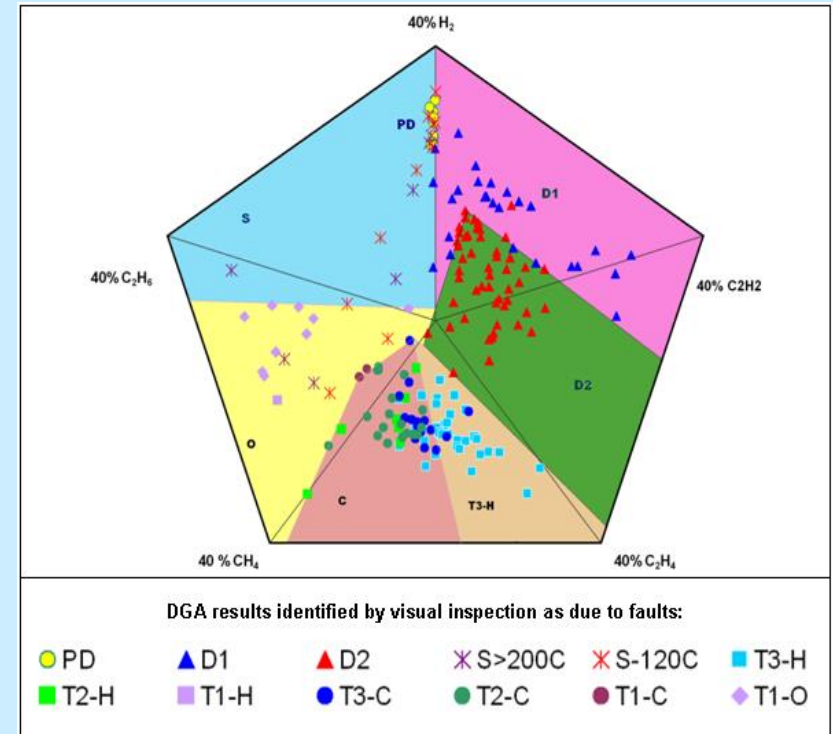
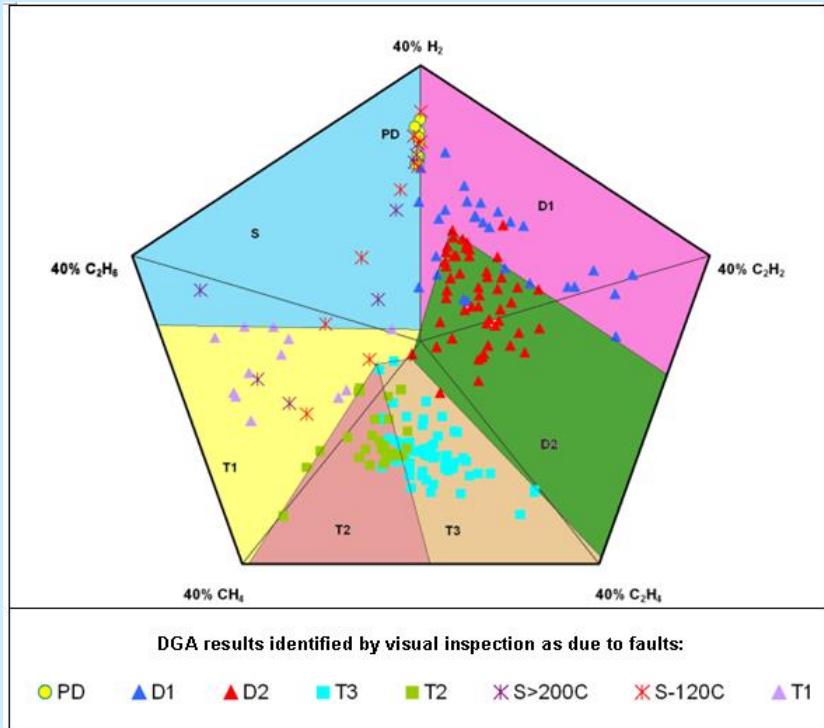


Triangle 5, using
 CH_4 , C_2H_4 and C_2H_6

Use of Triangles 4 and 5

- Triangle 4 should be used only with faults identified first with Triangle 1 as faults PD, T1 and T2.
- Triangle 5 should be used only with faults identified first with Triangle 1 as faults T2, T3.
- Triangles 4 and 5 should never be used for faults identified with Triangle 1 as electrical faults D1, D2 or DT.
- See excel file for algorithm for Triangles 1 to 7

Duval Pentagons 1 and 2 (2014)



Pentagon 1 and Pentagon 2,
using H_2 , C_2H_6 , CH_4 , C_2H_4 and C_2H_2

Use of Pentagons 1 and 2

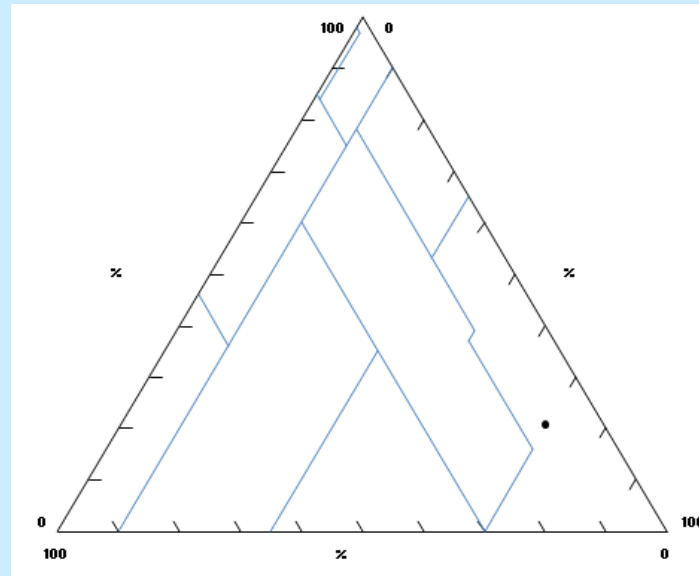
- Pentagon 1 can be used to identify the 6 basic types of faults (PD, D1, D2, T1, T2, T3) and stray gassing S.
- Pentagon 2 can be used to identify the 3 basic electrical faults D2, D1 and PD, and the 4 sub-types of faults S, O, C and T3/T2 in oil only (“T3-H, T2-H”).
- DGA in zones C indicate a possibility only of carbonization of paper, (with a probability of ~80%, not 100%).
- See excel file for algorithm for Pentagons 1 and 2

DGA Example 1

- Rectifier 20kV, 20 MVA, (A.M.Haug)

H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
555	1050	29	3520	489	327	4632	T3	-	T3-H	T3-H

(ppm)



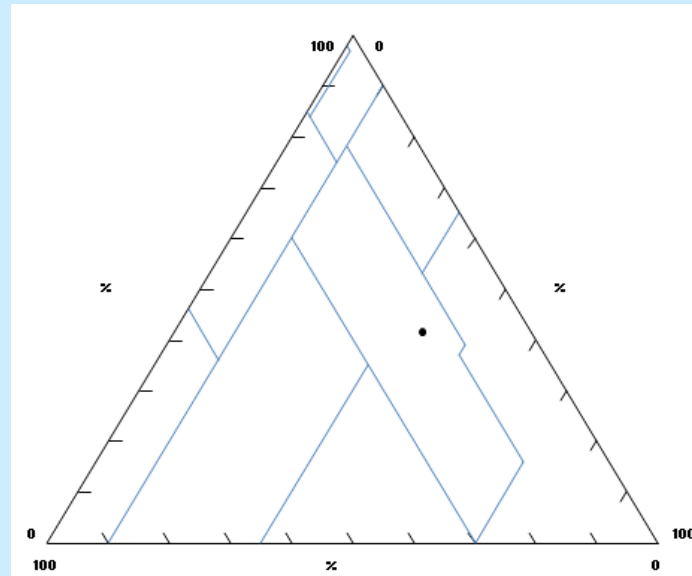
- Burnt selector found by inspection, as predicted (T3-H)

DGA Example 2

-Transformer 24kV, 40 MVA (R.Schneider)

H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
754	2647	6	2590	1127	105	538	T2	C	C	C

(ppm)



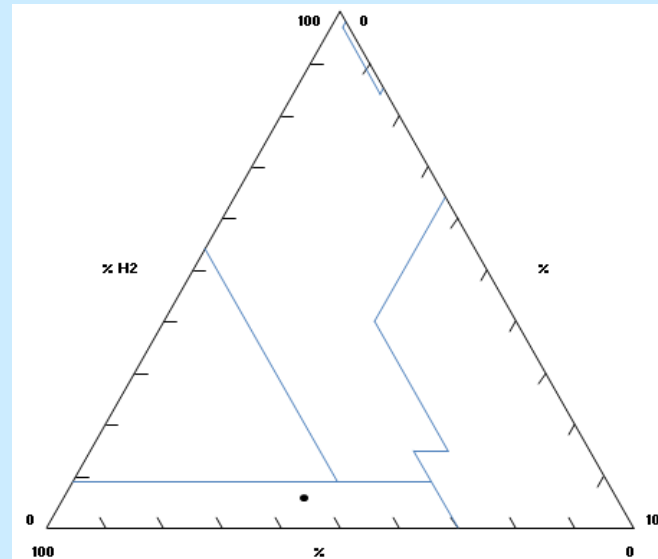
- Buchholz alarm, carbonized leads found by inspection (C)

DGA Example 3

- Transformer 66kV 25MVA (E. Alzieu)

H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
29	204	0	17	264	24	2000	T1	O	(O)	O

(ppm during one year)



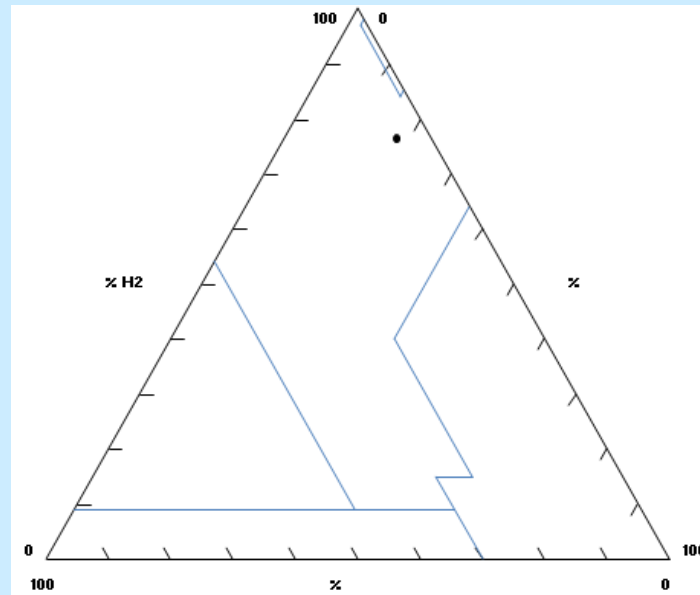
- Overheated insulated iron sheets of lower frame found by inspection (O)

DGA Example 4

- Sealed transformer with coated paper (S.Bhumiwat)

H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
48273	11485	7	8	3452	107	1943	T1	S	-	S

(ppm)



- Overheating of pressboard between windings found by inspection

Single vs. Multiple Faults in Transformers

- Triangles 1, 4, 5 and Pentagons 1, 2, as well as all other diagnosis methods (Rogers, etc), were initially developed for detecting single faults only.
- However, mixtures of faults (multiple faults) sometimes occur in service, and are more difficult to identify with certainty because they produce mixtures of gases.

Single vs. Multiple Faults in Transformers

- For instance, mixtures of faults D1 and T3 may wrongly appear in terms of gas formation in zone D2 of Triangle 1 and Pentagon 1.
- Mixtures of faults T3 in oil only (T3-H) and overheating O may appear wrongly in zone C of Triangle 5 and Pentagon 2.
- Mixtures of faults S and T2 may inaccurately appear in zones T1 or O of Pentagons 1 or 2.

Detection of Multiple Faults in Transformers

- When there is a change in gas pattern with time, and when Triangles 1, 4, 5 and Pentagons 1, 2 do not provide the same diagnosis, this may be an indication of multiple faults.
- This is because each graphical representation is more sensitive to some gases (e.g., H_2 or C_2H_4) and therefore to some faults (e.g., S or T3 in oil).
- To more easily detect multiple faults, it may be useful to subtract the last gas formation from the most recent one.

DGA Example 5 of Multiple Faults

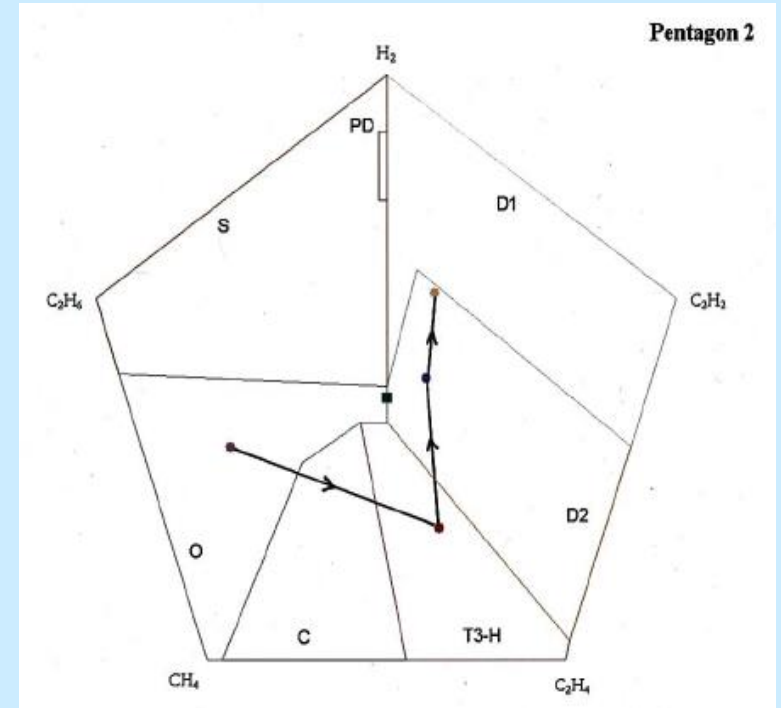
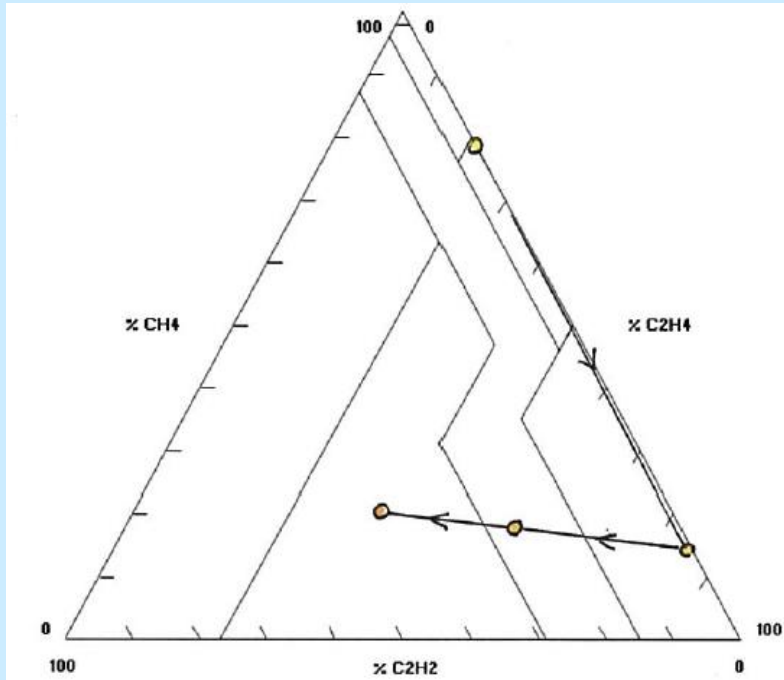
- Transformer 180kV 47MVA (E. Alzieu)

	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	Tr1	Tr4	Tr5	Pent2
2005	5	150	0	40	200	T1	O	-	O
2013	7	17	1	99	23	T3	-	T3-H	T3-H
2014	193	48	66	154	22	DT	-	-	D2
Delta	186	31	65	55	0	D2	-	-	D1

(ppm)

- Overheating of leads and windings and arcing in windings found by inspection

DGA Example 5



- A mixture of faults O + T3-H + D actually occurred in this transformer

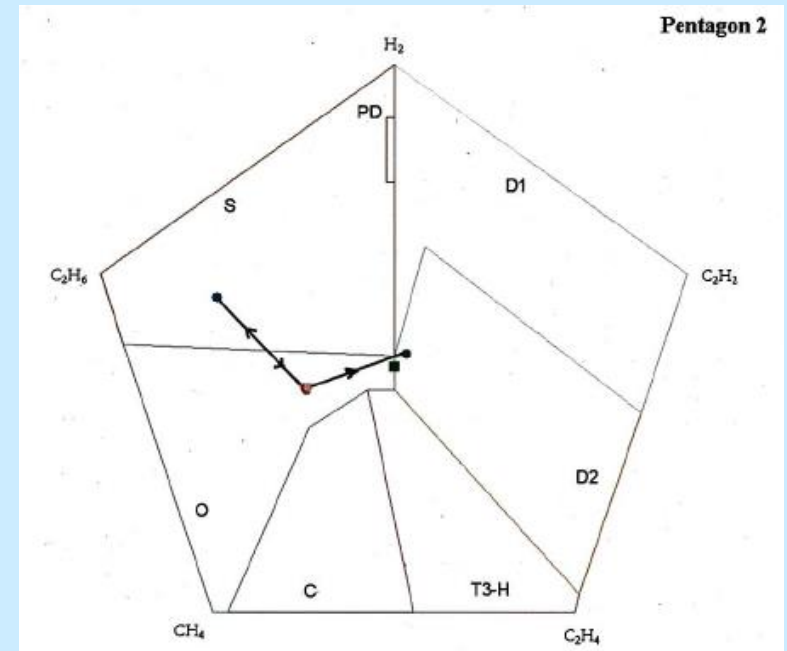
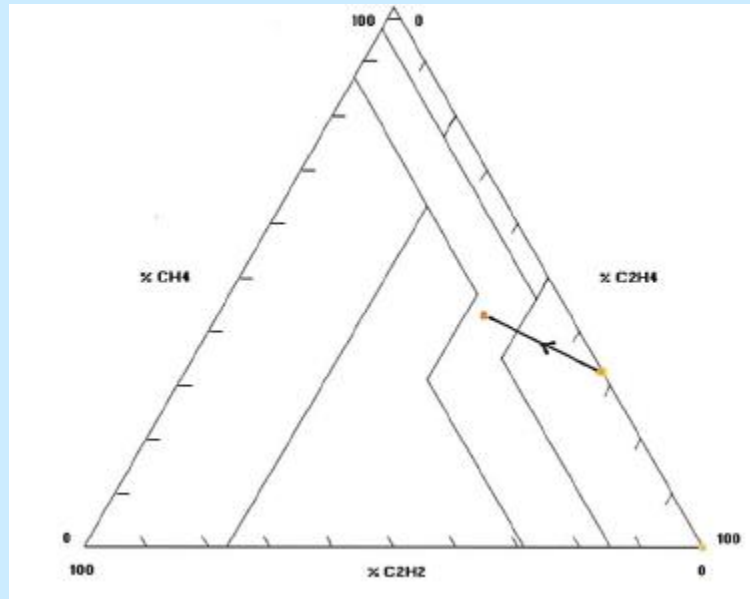
DGA Example 6

- HV Hydro transformer (E.Alzieu)

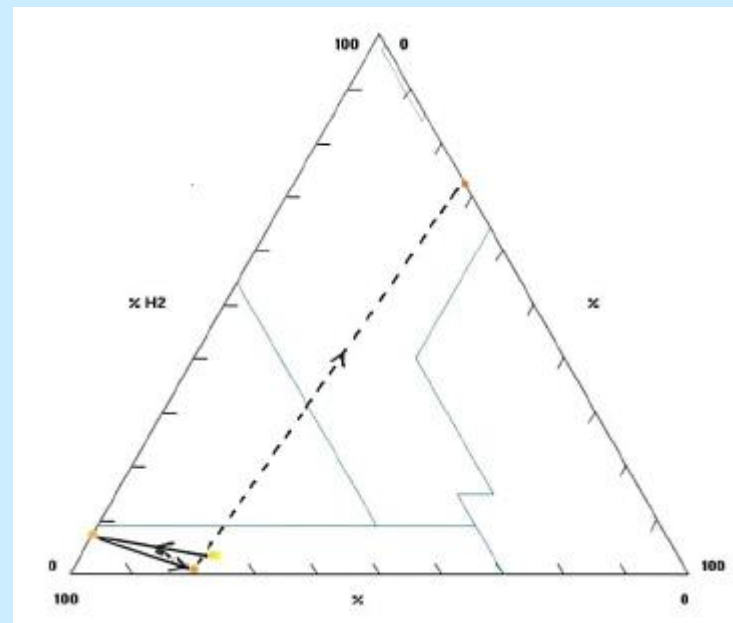
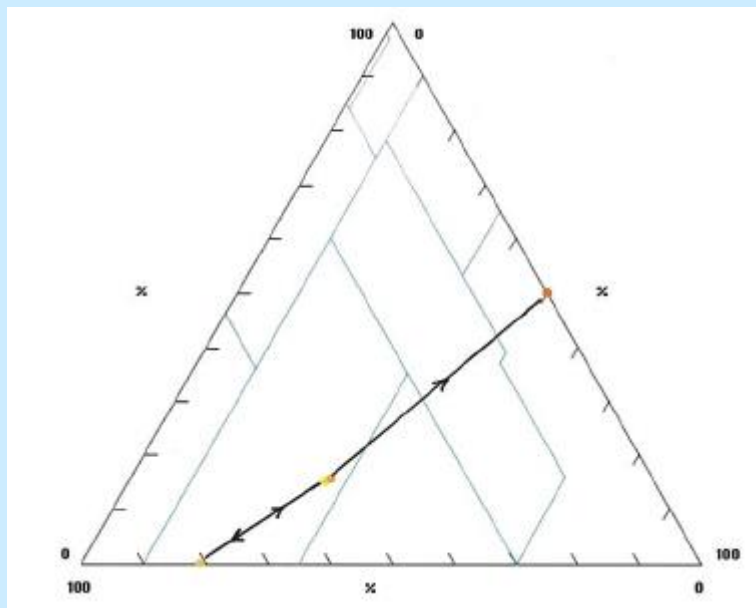
	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
<u>June 3</u>	21.5	140.3	0	286.8	473.9	150.6	8148	T3	(O)	T3/O	O
<u>June 12</u>	22.9	139.2	0	290.9	491.5	151	8224	T3	(O)	T3/O	O
Delta	1.4	0	0	4.1	17.6	0.4	76	T3	(O)	O	S
<u>June 16</u>	35.7	144.1	1.6	295.8	476.4	151.8	8416	T3	(O)	T3/O	O
Delta	12.8	4.9	1.6	4.9	0	0.8	196	DT	⊥	(T3-H)	D

(ppm)

DGA Example 6



DGA Example 6



- A mixture of faults S + O + T3-H + D appears to have occurred in this transformer.

DGA Example 7

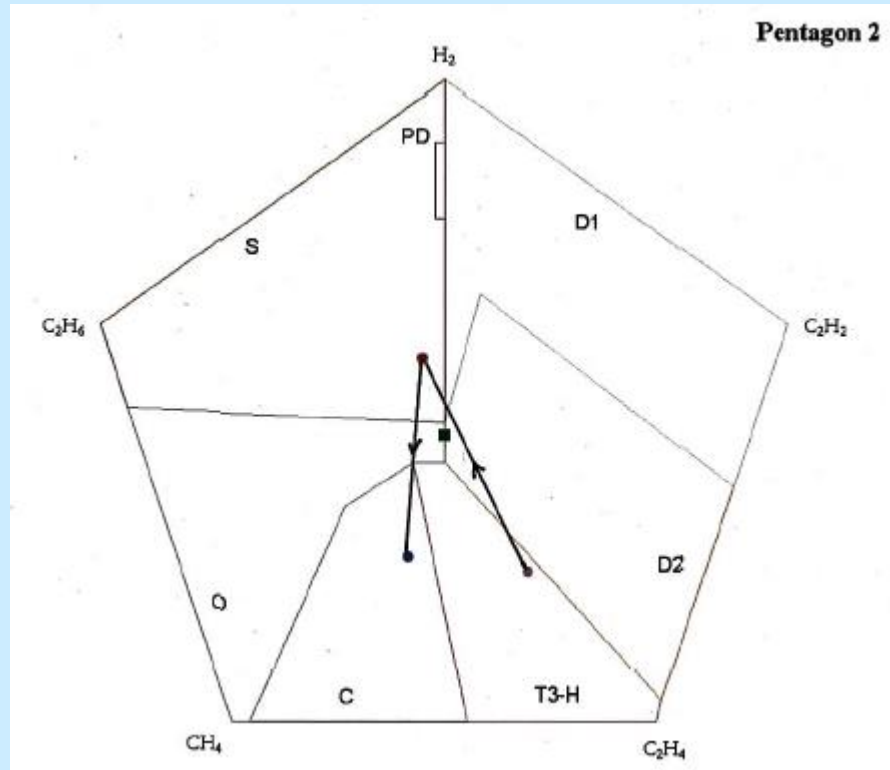
- Rectifier 120kV 48MVA (A.M. Haug)

	H ₂	CH ₄	C ₂ H ₂	C ₂ H ₄	C ₂ H ₆	CO	CO ₂	Tr1	Tr4	Tr5	Pent2
2010	13	6	0	69	7			T3	-	T3-H	T3-H
2010	5170	1674	0	380	195	554	5769	T1	S	(T2-H)	S
	(80)	(800)	0	380	195			(T2)	(C)	(C)	(C)
2013	223	2121	0	2883	1708	338	3708	T3	-	C	C

(ppm)

- Carbonization of paper in windings found by inspection in 2010, and carbonization of leads in 2013.

DGA Example 7

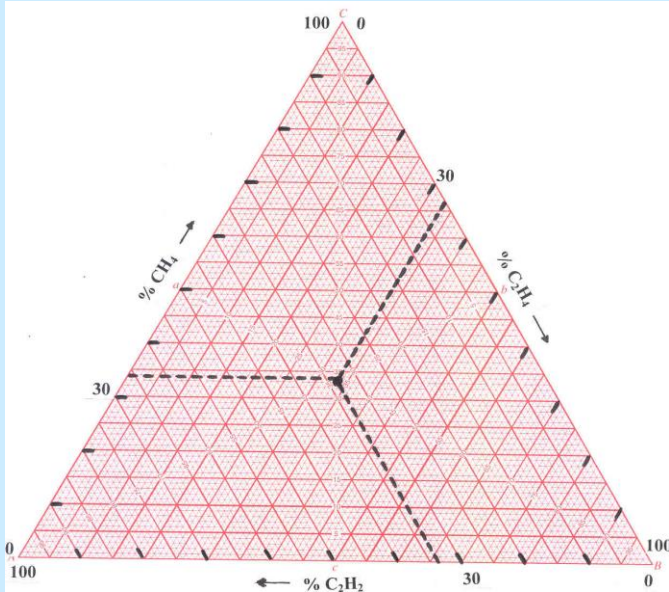


- A mixture of faults S and C actually occurred in 2010

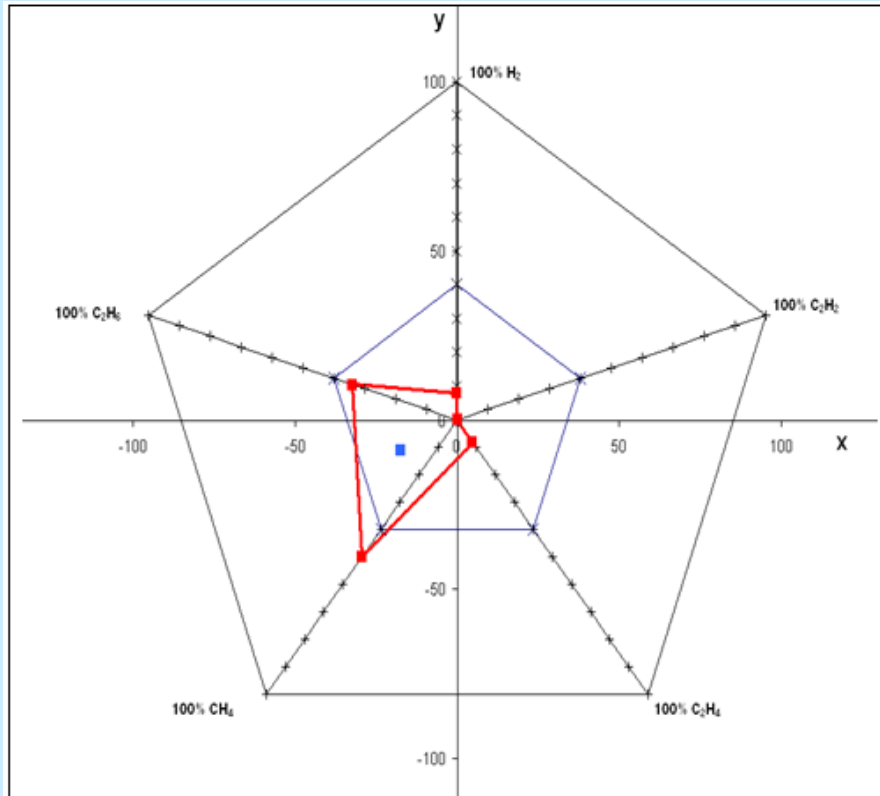
Use of Triangular Coordinates

Example:

If $\text{CH}_4 = \text{C}_2\text{H}_4 = \text{C}_2\text{H}_2 = 100 \text{ ppm}$,
the triangular coordinates to use are:
 $\% \text{CH}_4 = \% \text{C}_2\text{H}_4 = \% \text{C}_2\text{H}_2 = 100 / 300 = 33.3\%$,
and correspond to only one point in the Triangle.



Calculation of DGA Coordinates in the Pentagons



- DGA results used in this example for H_2 , C_2H_6 , CH_4 , C_2H_4 and C_2H_2 : 31, 130, 192, 31 and 0 ppm,
- i.e., a relative percentage for each gas of 8, 34, 50, 8 and 0%, respectively.
- The DGA point is the centroid (blue square) of the red polygon.
- All DGA points fall inside the blue (40%) pentagon.

Use of Duval Triangles and Pentagons for the Interpretation of DGA in Transformers

QUESTIONS?

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